

MIRAMAR MCAS, CA, USA

WMO: 722931

Lat: 32.867N

Lon: 117.133W

Elev: 145

StdP: 99.59

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 93107

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	4.0	5.4	-12.2	1.3	16.8	-9.3	1.7	17.1	8.5	15.1	7.5	15.2	2.5	80	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.9	33.4	18.8	31.4	18.7	29.6	18.6	22.0	29.0	21.3	27.9	20.7	26.9	3.8	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	14.7	24.8	19.0	14.0	24.2	18.3	13.4	23.4	65.0	29.3	62.4	28.1	60.1	27.0	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.0	5.7	5.2	1.7	37.6	1.4	2.7	0.6	39.6	-0.2	41.2	-1.0	42.7	-2.0	44.7		
(5)			-1.0	23.6	1.4	1.0	-2.0	24.3	-2.9	24.9	-3.7	25.5	-4.7	26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.7	13.9	13.9	14.7	15.8	17.4	19.6	22.0	22.9	22.1	19.4	16.3	13.5
	DBStd	4.28	2.95	3.03	2.73	2.88	2.31	2.16	2.22	2.24	2.87	3.09	3.17	2.62
	HDD10.0	8	3	2	1	1	0	0	0	0	0	0	1	2
	HDD18.3	774	139	128	117	84	44	9	1	0	2	23	77	151
	CDD10.0	2803	124	112	147	176	231	287	373	398	362	291	190	112
	CDD18.3	527	3	4	4	10	17	47	116	140	114	55	16	2
	CDH23.3	3964	74	81	85	164	146	260	643	919	817	512	224	38
	CDH26.7	1307	11	19	21	53	52	71	172	301	312	213	77	6
Wind	WSAvg	2.4	2.5	2.6	2.5	2.6	2.5	2.3	2.2	2.2	2.3	2.4	2.4	2.4
Precipitation	PrecAvg	269	51	51	50	23	5	2	1	1	5	9	34	35
	PrecMax	714	225	300	196	98	32	19	12	36	32	67	144	119
	PrecMin	62	0	0	0	0	0	0	0	0	0	0	0	1
	PrecStd	150	54	61	50	26	7	4	2	6	7	15	35	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.0	29.3	29.1	32.0	32.0	32.3	33.4	34.6	36.2	35.7	32.5	27.3
		MCWB	13.3	14.0	14.4	14.7	15.3	19.6	21.4	21.1	20.5	17.3	15.0	12.2
	2%	DB	25.4	26.0	25.5	27.6	27.2	28.5	30.3	32.0	32.6	32.0	28.6	23.7
		MCWB	12.4	13.0	14.0	14.5	17.0	19.4	20.7	20.8	20.4	16.4	13.7	12.0
	5%	DB	23.0	22.8	22.8	24.6	23.9	26.1	28.5	29.8	30.2	28.7	25.7	21.6
		MCWB	12.0	12.4	13.6	14.3	16.9	18.7	20.3	20.4	19.9	15.8	13.1	11.5
	10%	DB	20.4	20.1	20.5	22.0	22.1	24.2	26.9	27.9	27.8	25.8	22.8	19.3
		MCWB	11.6	12.6	13.5	14.4	16.0	17.8	19.9	20.3	19.7	16.5	13.4	11.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.4	16.6	17.5	17.7	20.0	20.8	22.8	23.0	23.1	20.8	17.5	16.4
		MCDB	21.5	22.1	22.5	24.9	25.9	28.9	29.8	31.5	31.5	26.8	22.3	20.0
	2%	WB	15.2	15.3	15.9	16.5	18.2	19.8	21.7	21.9	21.9	19.4	16.7	15.2
		MCDB	19.7	21.1	21.6	23.6	24.2	27.0	28.0	29.1	28.9	26.1	22.0	18.7
	5%	WB	14.1	14.4	15.1	15.6	17.1	19.0	20.9	21.2	21.1	18.4	15.8	14.4
		MCDB	18.0	19.5	20.1	21.8	22.5	25.4	26.9	28.0	27.5	24.0	21.1	17.8
	10%	WB	13.3	13.5	14.2	14.8	16.4	18.2	20.2	20.5	20.3	17.7	15.1	13.5
		MCDB	17.5	18.3	18.6	20.3	21.0	23.7	25.8	26.8	26.1	22.8	20.2	16.9
Mean Daily Temperature Range	5% DB	MDBR	12.0	11.3	10.7	10.7	8.9	9.1	9.3	9.9	10.4	11.3	12.3	11.7
		MCDBR	16.3	16.6	15.4	16.7	13.1	12.7	11.8	12.6	14.2	17.3	17.0	15.7
		MCWBR	7.8	7.8	7.6	7.8	6.6	5.7	4.7	5.1	5.7	7.9	8.0	8.2
	5% WB	MCDBR	11.2	12.0	11.6	12.6	11.1	11.6	10.7	11.1	11.6	11.7	12.5	10.2
		MCWBR	6.4	6.7	6.5	6.5	5.8	5.4	4.5	4.8	5.2	5.8	6.8	6.2
Clear-Sky Solar Irradiance	taub	0.310	0.320	0.333	0.344	0.363	0.362	0.391	0.391	0.358	0.364	0.318	0.321	
	taud	2.558	2.543	2.490	2.465	2.433	2.438	2.437	2.416	2.490	2.443	2.564	2.555	
	Ebn at Noon	915	939	947	944	923	918	890	887	908	874	898	880	
	Edh at Noon	81	91	104	111	116	115	114	115	102	99	80	78	
All-Sky Solar Radiation	RadAvg	3.31	4.17	5.49	6.63	7.00	7.33	7.14	6.86	5.92	4.61	3.60	2.99	
	RadStd	0.21	0.36	0.30	0.35	0.50	0.44	0.45	0.28	0.30	0.33	0.23	0.21	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.94	N/A	N/A	+1.41	N/A	N/A	N/A	-197	N/A	N/A
(47) Regional (41 neighbors)		+0.66	N/A	-1.61	+0.75	N/A	N/A	N/A	-107	+235	+124

Nomenclature: See separate page